Agnel Charities

Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

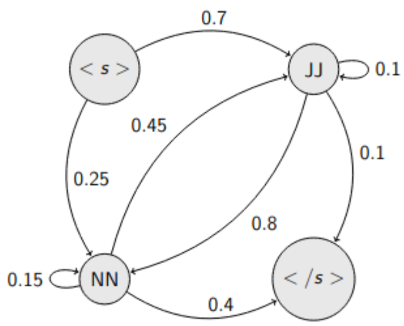
Department of Computer Engineering

Sem : VII	**Department of Computer Engineering**	**Time: 1 Hour**
Subject : Natural Language Processing	**Internal Assessment-I (SH-2025)**	**Total Marks: 20**

Note :

- **ALL** questions are **compulsory.**
- Assume suitable data wherever required, but justify the same.

Q. 1		Attempt **any two** (Each question carries 02 marks)	CO	BTL
	a)	With the given HMM, calculate the probability that the sentence **“hot lead cold feet”** will be assigned the parts of speech; **JJ NN JJ NN** Values on the edges are transition probabilities and the table below gives emission probabilities:	3	3



	hot	lead	cold	feet
JJ	0.4	0.1	0.45	0.05
NN	0.01	0.55	0.02	0.42

			CO	BTL
	b)	Compute the probability of the parse tree of the input string “**The mouse chased the cat”,** given the following partial PCFG	3	3

S ⟶ NP VP 0.75

NP ⟶ DT NN 0.25

VP ⟶ VB NP 0.35

NN ⟶ **mouse** 0.25 | **cat** 0.1 | **dog** 0.4

VB ⟶ **chased** 0.3 | **dodged** 0.5

DT ⟶ **the** 0.2 | **a** 0.35

Agnel Charities

Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

Department of Computer Engineering

	c)	Perform left most derivation and compute the parse tree to check the string **“The Panda eats leaves”**, belongs to the following grammar: **S ⟶ NP VP** **NP ⟶ DT NN** **VP ⟶ VB NN** **DT ⟶ a | the** **NN ⟶ Man | Woman | Panda | Lion | leaves** **VB ⟶ eats | shoots | leaves**	3	3
		Attempt **any three** (Each question carries 02 marks)		
	d)	Given the following regular expression R = “\b[A-Za-z]*[^aeiou]{2}[a-z]*\b” and the following input sentence, **write** all words matched with regular expression R **Dolly on a trolley got a seat by golly. Guss on a bus got a seat without fuss.**	2	3
	e)	**Write** the inflectional morphemes identified in the following discourse. **Them that asked no questions, cleverer not told lies.**	2	3
	f)	Compute the perplexity of “I like sweets”, by considering the bigram model from the following text. I like many things to eat. I like cake. I like biscuits. And, yes, sweets. I do like them. I like jam. It is like sweets.	2	3
	g)	Compute the measure of the following words, with respect to Porter Stemmer: **Pyrotechnique, acronymania, gyroscope, pylorus**	2	3
Q. 2		Attempt **any one** (Each question carries 05 marks)		
	a)	Consider the following corpus • <s> the/DT students/NN pass/V the/DT test/NN </s> • <s> the/DT students/NN wait/V for/P the/DT result/NN </s> • <s> teachers/NN test/V students/NN </s> Compute the emission and transition probabilities for a bigram HMM.	3	3
	b)	Consider the grammar of Q. 1. (c) and compute the CKY table to check the string “The woman shoots Lion”, belongs to the given grammar.	3	3
Q. 3		Attempt **any one** (Each question carries 05 marks)		
	a)	Exemplify the challenges in natural language processing.	1	2
	b)	Summarize three applications of NLP	1	2